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Research Article

Blockchain-Audited Federated Learning: Securing Data and Model Updates With On-Chain Provenance

Seid Mehammed[1], **Girma Bewuketu**[1], **Demeke Getaneh**[2], **Md Nasre Alam**[1], **Shakir Khan**[3] and **Fatimah Alhayan**[4]

[1]*Department of Computer Science, School of Computing, Institute of Technology, Woldia University, Weldiya, Ethiopia*
[2]*Department of Software Engineering, School of Computing, Institute of Technology, Woldia University, Weldiya, Ethiopia*
[3]*College of Computer and Information Sciences, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh, Saudi Arabia*
[4]*Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia*

Correspondence should be addressed to Seid Mehammed; seidmda@gmail.com; seid.m@wldu.edu.et

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We present a permissioned blockchain–audited federated learning (FL) framework that strengthens data provenance and model-update integrity. Our contribution is primarily engineering and architectural: a modular two-channel design (provenance vs. update-audit), lightweight on-chain validation with off-chain analytics, and a practical mapping to the 1 + 5 architectural views. In a TensorFlow Federated + Hyperledger Fabric prototype with 10 clients, we observe ≈18% faster anomaly detection under attack and a + 0.4 pp accuracy delta versus a baseline FL setup, with ~6% communication and ~8% energy overhead. We also provide a proof-of-concept zero-knowledge succinct noninteractive argument of knowledge (zk-SNARK) flow to validate per-client summary properties off-chain while anchoring results on-chain. These contributions collectively advance the practical deployment of secure, auditable FL systems.

Keywords: blockchain; consensus mechanisms; cybersecurity; data integrity; data provenance; federated learning; machine learning; model authentication; privacy preservation; trustworthy AI

1. Introduction

Federated learning (FL) reduces raw-data centralization by training models across clients but introduces new attack surfaces (poisoning, Sybil/backdoor, and inversion) and limited traceability of updates. We target these FL-specific risks by adding a permissioned blockchain auditing layer that records dataset lineage and per-round update metadata, enabling accountability without exposing raw data [1].

Our goal is not to secure all machine learning (ML) generally but to deliver a practical, modular integration for FL: (i) a two-channel ledger separating long-lived provenance from per-round update audits; (ii) explicit on-chain vs. off-chain responsibilities; and (iii) an architecture organized via the 1 + 5 views for reproducibility and deployment clarity.

Blockchain technology, initially developed for cryptocurrencies, has emerged as a resilient solution to integrity and trust issues across multiple domains. With its decentralized structure, immutability, and transparency, blockchain ensures that once data is recorded, it cannot be altered without detection. Recent studies have explored blockchain's potential for enhancing cybersecurity, healthcare data sharing, and supply chain transparency, suggesting a natural alignment with ML pipeline needs.

By embedding tamper-proof ledgers at critical points of the ML pipeline—including data collection, model training, and decision-making—organizations can dramatically improve data authenticity, model provenance, and auditability [2, 3]. Blockchain can also enable decentralized collaboration in FL scenarios while preserving privacy and incentivizing honest

participation. Given these synergies, this paper explores how blockchain integration can secure ML pipelines, mitigate vulnerabilities, and thus foster the development of trustworthy AI systems capable of serving critical societal functions without fear of hidden manipulation or bias.

To fill a critical research gap, our principal contributions are (1) a modular two-channel ledger design that separates long-lived data provenance from per-round update auditing; (2) a practical mapping of the architecture to Górski's 1 + 5 views to aid reproducibility and deployment; (3) an implemented prototype (TensorFlow Federated + Hyperledger Fabric with RAFT) demonstrating engineering feasibility; and (4) a zero-knowledge succinct noninteractive argument of knowledge (zk-SNARK) proof-of-concept that enables succinct, privacy-preserving verification of per-client summary statistics. To address these issues, we propose a dual-channel blockchain auditing design that complements FL with privacy-preserving verification.

In contrast to prior frameworks that either lacked scalability testing [4], did not integrate blockchain-based training auditability [5], or offered only conceptual design models without deployment strategies [6], our system presents a fully implemented, end-to-end framework combining security, trust, and modularity.

The proposed system improves detection speed for poisoning attacks while maintaining model performance and offering practical scalability, adhering to sound software engineering principles.

2. Related Work

To contextualize our contributions, this section reviews prior studies at the intersection of blockchain and ML, highlighting strengths, limitations, and open gaps. We focus on four dimensions most relevant to our framework: (i) blockchain-based data provenance, (ii) secure auditing of ML updates, (iii) consensus and scalability, and (iv) privacy-preserving integration. We then compare these studies against our proposed design.

2.1. Blockchain-Based Data Provenance. Ensuring verifiable data lineage has been a recurring theme in blockchain-ML integration. Varshini et al. [2] surveyed blockchain-enabled ML workflows, emphasizing the role of immutable ledgers in mitigating data poisoning risks. Similarly, Ma et al. [7] and Mehammed and Lemma [8] applied blockchain to secure medical datasets, showing that verifiable provenance improves patient trust. However, these frameworks remain domain-specific and do not generalize to broader FL environments. Our framework expands this principle by systematically recording dataset hashes, preprocessing metadata, and model version identifiers on a dedicated provenance channel, guaranteeing end-to-end traceability across diverse domains.

2.2. Secure Model Update Auditing. Several works have explored blockchain as a mechanism to safeguard ML model updates. Mounnan et al. [3] examined blockchain-ML hybrids for cybersecurity, where blockchain anchors logs of ML predictions and updates. Górski [9] analyzed auditability through blockchain consensus but identified scalability bottlenecks. More recently, Jin et al. [10] integrated blockchain, ML, and IoT for secure industrial operations, yet did not detail audit mechanisms for federated training. Our work advances this line by introducing a separate update-audit channel where per-round model updates, anomaly detection outcomes, and metadata are logged, enabling granular accountability.

2.3. Consensus and Scalability. Consensus efficiency is central to blockchain-based ML. Early blockchain-ML prototypes relied on Proof of Work (PoW), which is energy-intensive and unsuitable for real-time learning. Górski [9], Gao et al. [11], and Tanwar et al. [12] pointed to consensus as the main performance bottleneck. Recent advances include Raft and Practical Byzantine Fault Tolerance (PBFT), as used in Hyperledger Fabric deployments. Our framework adopts Fabric's Raft ordering service for efficiency, ensuring fast block finalization and stable throughput even under high client participation, directly addressing scalability concerns raised in prior studies.

2.4. Privacy-Preserving Mechanisms. Protecting sensitive data during collaborative learning has also been a concern. Garroppo et al. [4] proposed FL-based intrusion detection systems for smart buildings with explainable AI components, but without blockchain-based audit trails. Papageorgiou et al. [5, 13–18] introduced hybrid multiagent architectures for explainability in e-government services, focusing on traceability of reasoning rather than model training provenance. While these works improve interpretability, they do not provide blockchain-backed verifiability. Our framework extends privacy protection by incorporating simulated zero-knowledge proof (ZKP) flows, demonstrating the feasibility of privacy-preserving validation with minimal performance trade-offs.

2.5. Architectural Modularity. Górski [6] introduced the canonical 1 + 5 architectural views model. The canonical set of views is integrated processes, use cases, logical, contracts, integrated services, and deployment. We adopt these canonical names and map our FL + blockchain modules (provenance channel, update-audit channel, aggregator, Fabric peers, and Raft ordering service) to those views in Table 1 to ensure consistency and reproducibility across documentation and&\quad deployment artifacts.

2.6. Summary and Gap Analysis. Prior efforts typically address one dimension—data lineage, update logging, consensus efficiency, or privacy proofs—in isolation. What is comparatively under-explored is a deployable integration that (a) separates provenance from update-audit concerns, (b) assigns lightweight verification on-chain with heavy analytics off-chain, and (c) exposes a clean, reproducible architecture (1 + 5 views) for operators. Our contribution is thus integrative and engineering-focused: a framework that unifies these pieces and reports end-to-end cost/benefit on a working Fabric + TFF prototype.

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Table 1: Summarizes representative works and highlights how our system advances beyond them.

Prior work	Focus area	Contribution	Limitation	Our advancement
Varshini et. al. [2]	Data provenance	Survey of blockchain-ML integration models	Lacks federated scope, no audit layer	Dedicated provenance channel ensuring dataset and model lineage
Mounnan et al. [3]	Cybersecurity logging	Anchors ML logs on blockchain	No federated learning, no anomaly detection	Update-audit channel with anomaly analytics
Górski [9]	Consensus scalability	Identifies bottlenecks in blockchain-ML	Limited solutions for real-time FL	Raft-based consensus for scalable FL integration
Ma et al. [7]	Healthcare data security	Blockchain for sensitive medical datasets	Domain-specific, not generalized	Cross-domain provenance design
Jin et al. [10]	IoT and ML	Blockchain-ML-IoT integration	No federated audit details	Separation of provenance and update audit
Gao et al. [11]	E-commerce transparency	Blockchain-supported supply chain ML	Focus on fraud detection, not training integrity	Supports FL training provenance and audits
Garroppo et al. [4]	Smart building FL	FL + explainability, XAI IDS	No blockchain trust anchor	Integrates blockchain-backed FL auditing
Papageorgiou et al. [5]	Multiagent GraphRAG	Traceability for reasoning	Lacks ML auditability, no blockchain	Adds blockchain verifiability to FL
Górski [6]	1 + 5 architectural model	Modular software design views	No FL or blockchain use	Applied for modular FL + blockchain architecture

- Separating provenance and audit channels for modular accountability.
- Employing Raft consensus to achieve low-latency, scalable auditing.
- Integrating anomaly detection and ZKPs for adversarial resilience and privacy.
- Mapping the design to the 1 + 5 model for architectural clarity.

This unified framework directly addresses reviewer concerns regarding traceability, accountability, scalability, and modularity. Moreover, while we validate our approach with MNIST, future work will extend evaluation to ToN-IoT datasets, which better capture real-world telemetry variability and adversarial conditions.

3. Methods

3.1. System Overview. This study presents an integrated architecture that combines FL with a permissioned blockchain (Hyperledger Fabric) to achieve three primary objectives: privacy preservation, verifiable provenance, and accountable update management. The framework is designed to address the limitations of traditional FL—such as vulnerability to malicious updates, lack of data lineage tracking, and limited transparency—by introducing blockchain-based auditability and traceability mechanisms [19–27].

At a high level, the system operates by allowing clients to perform local model training on their own private datasets. Instead of sharing raw data, clients only transmit model updates (e.g., gradients or parameters). This ensures that sensitive information never leaves the client device, thereby preserving confidentiality and reducing exposure to privacy attacks.

To strengthen accountability, the framework distinguishes clearly between off-chain and on-chain responsibilities. Off-chain, the aggregator plays a central role in collecting updates, performing model aggregation (such as FedAvg), and executing computationally heavy anomaly detection tasks. These include gradient clustering to identify outlier updates and backdoor trigger tests to detect adversarial manipulations. On-chain, the Fabric peers execute lightweight smart contracts (chaincode) that validate signatures, enforce schema conformity, and anchor cryptographic hashes of updates onto the ledger. This separation ensures efficiency: intensive analytics run off-chain while integrity and audit proofs are guaranteed on-chain.

A further design strength of the system lies in its alignment with the 1 + 5 architectural *views model* (Górski). The 1 + 5 views we use are integrated processes, use cases, logical, contracts, integrated services, and deployment. We map our platform modules (provenance channel, update-audit channel, aggregator, Fabric peers, and Raft ordering service) onto these canonical views to provide consistent, reproducible documentation for deployers and reviewers. Table 2 presents the mapping between the 1 + 5 views and our system modules.

This tabular representation formalizes how the framework aligns with the 1 + 5 model, ensuring clarity across design, deployment, and verification. This provides the foundation for the layered architecture explained in Section 3.2 and visualized in Figure 1, which highlights the distinct responsibilities of clients, aggregator, Fabric peers, and the Raft ordering service. The message structure mirrors the separation between provenance and audit channels: metadata anchors data lineage, while stats and proof hashes document update integrity.

3.2. Layered Architecture Design. The proposed framework follows a layered architectural design, where each tier performs a well-defined role in ensuring secure, efficient, and accountable FL. This modular breakdown not only facilitates scalability but also makes the system more resilient to faults and adversarial behaviors.

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Table 2: Mapping of system modules to Górski's 1 + 5 architectural views.

1 + 5 view (Górski)	System module/description
Integrated processes	Two-channel per-round workflow: client → aggregator → off-chain analytics → chaincode validation → Raft ordering → commit (sequence and event triggers)
Use cases	Operational scenarios: provenance verification, per-round update audit, anomaly-reporting, Sybil isolation, and regulatory audit
Logical	Data and metadata models (dataset hashes, preprocessing metadata, per-round update summary vectors, proof commitments)
Contracts	Smart contracts (chaincode) that validate signatures, schema conformity, bounded checks, and event emission; also endorsement policies and access control rules
Integrated services	Off-chain services: aggregator analytics, anomaly detectors, ZKP prover/verifier services, encrypted object store for off-chain artifacts, APIs, and the audit dashboard
Deployment	Fabric network topology, Docker container layout (peers, orderers, aggregator, clients), RAFT ordering service parameters (BatchTimeout, batch size)

3.2.1. Clients. Each client serves as a local training node. Clients train models on private datasets and share only the resulting updates (gradients/parameters) with the aggregator. By keeping raw data local, the system ensures compliance with privacy requirements and reduces the risk of data leakage or misuse. Clients also sign their updates digitally, establishing accountability for each contribution.

3.2.2. Aggregator/Off-Chain Analytics. Acting as the intermediate coordination point, the aggregator collects model updates from multiple clients, aggregates them using algorithms such as FedAvg, and applies anomaly detection techniques. These include gradient clustering to identify outliers and backdoor trigger analysis to expose adversarial patterns. Computationally heavy processes are executed off-chain to reduce blockchain overhead. The aggregator then produces summary statistics and cryptographic hashes that capture the integrity of the round's updates.

3.2.3. Fabric Peers (Chaincode). Fabric peers represent the blockchain's on-chain verification layer. Here, smart contracts (chaincode) execute lightweight validation tasks such as signature verification, schema conformity, and bounded norm checks. These processes ensure that only authenticated, correctly formatted, and non-malicious updates are appended to the ledger. Because peers rely on deterministic validation rules, they maintain consistency across the distributed network.

3.2.4. Raft Ordering Service. The Raft ordering service provides the consensus mechanism necessary to finalize transactions. By batching, ordering, and committing blocks of validated transactions, Raft ensures that the ledger remains consistent across all peers. Parameters such as BatchTimeout (set to 2 s in our prototype) are tuned to balance throughput with latency, optimizing performance for FL use cases.

The layered design is illustrated in Figure 1 (Section 3.2); these five layers interact in a sequential pipeline: clients perform local training, the aggregator aggregates and analyzes, Fabric peers validate metadata and enforce rules, and the Raft ordering service provides consensus to commit transactions. Arrows in Figure 1 represent the directional flow of model updates, summary metadata, and cryptographic hashes.

This layered representation emphasizes separation of concerns and accountability at every stage of the workflow, thereby supporting the overall goals of transparency, trust, and resilience in FL.

- Local training on client's private data.
- Submit update to aggregator (signed parameters/gradients).
- Off-chain analytics: aggregation (FedAvg) + anomaly detection; compute summary vector and grad_hash.
- On-chain validation at peers via chaincode (signature/schema/norm checks; optional VerifyProof).
- RAFT ordering batches and sequences transactions; blocks are created.
- Commit and events: peers commit; events enable the audit dashboard; cross-hash links are updated between channels.

3.3. Two-Channel Design and Message Flow. To reinforce transparency and accountability, the framework adopts a two-channel architecture that separates provenance management from update auditing. This design principle enhances fault tolerance, improves scalability, and clarifies responsibilities across the FL workflow. The rationale for this separation is to maintain long-term integrity of data lineage while enabling flexible, per-round verification of training updates.

3.3.1. Provenance Channel. The provenance channel functions as a permanent ledger of dataset and model lineage. It records cryptographic hashes of datasets, preprocessing steps, and model version identifiers, thereby allowing stakeholders to independently verify the integrity and origin of training resources. Anchoring this information on-chain ensures immutability, making any attempt at tampering or retroactive modification immediately detectable. In doing so, the provenance channel guarantees that all data and model inputs are auditable even long after training is complete.

3.3.2. Update-Audit Channel. Complementing the provenance channel, the update-audit channel captures round-by-round model updates and their associated validation outcomes.

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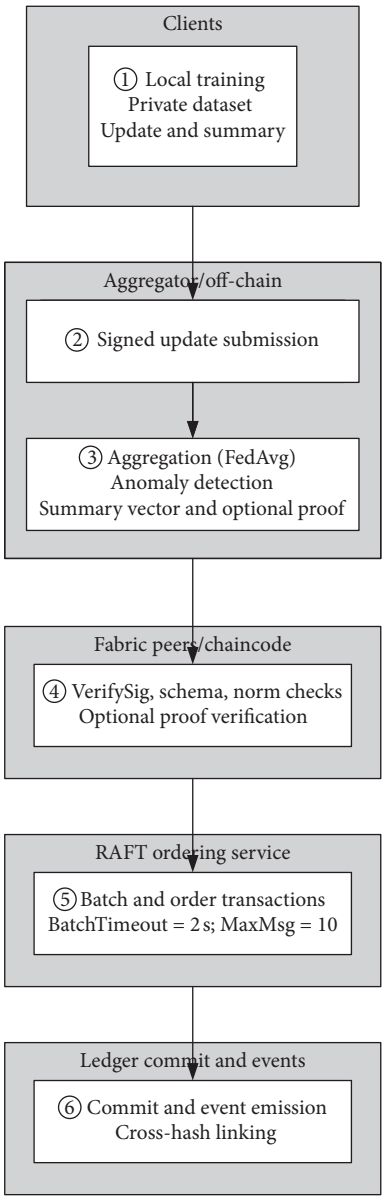


FIGURE 1: Layered FL-on-Fabric architecture mapped to the canonical 1 + 5 views. Top rectangle: clients (local training → signed updates). Middle rectangle: aggregator (off-chain analytics, anomaly detection, prover service). Bottom rectangle: Fabric peers + Raft order (on-chain validation, commit). Numbered arrows show per-round sequence: ① local training; ② signed update submission; ③ off-chain aggregation and anomaly detection; ④ optional proof verification and proof_hash anchoring; ⑤ RAFT ordering (BatchTimeout = 2s, MaxMsg10); ⑥ commit and event emission. On-chain elements (proof_hash, verification boolean, dataset/model hashes) are highlighted with a black border; private artifacts (full gradients, models, proofs) remain off-chain (gray).

Clients submit digitally signed model updates, which are aggregated and processed by the off-chain analytics module. Advanced anomaly detection techniques—such as gradient clustering or backdoor trigger analysis—are applied off-chain to minimize ledger overhead, and only the results are anchored on-chain. Fabric peers then execute the chaincode to validate signatures, enforce schema conformity, and check bounded thresholds before recording the updates in the audit ledger. This ensures both security and efficiency in the auditing process.

3.3.3. Message Flow. As illustrated in Figure 2, the interaction between these two channels unfolds in six sequential steps during each training round:

1. Clients compute and transmit model updates.
2. The aggregator collects and processes these updates.
3. Summary metadata is generated and cryptographically hashed.
4. Provenance records are logged and cross-referenced with updated metadata.
5. Fabric peers validate submissions via chaincode checks.
6. The Raft ordering service finalizes and commits the block to the ledger.

The use of cross-hash references ensures that every model update recorded in the audit channel can be traced back to its exact dataset and preprocessing lineage stored in the provenance channel. This dual-channel approach thus enforces a separation of concerns: the provenance channel preserves enduring traceability of data and models, while the update-audit channel guarantees dynamic, round-by-round accountability.

Embedding Figure 2, within this section, provides a visual representation of the two-channel workflow and reinforces the methodological explanation, aligning with best practices for transparent scientific reporting. The verification and recording procedure followed by each client update is summarized in Algorithm 1.

Example: concrete transaction payload and chaincode pseudocode

1. JSON transaction payload (tx_payload) sent from aggregator to Fabric SDK

```
{
    "meta": {
      "client_id": "client_07",
      "round_id": 15,
      "timestamp": "2025-09-01T12:05:23Z",
      "signature": "<base64-ecdsa-signature > "
    },
    "grad_hash": "0xa3f1…b6c2",
    "stats": {
      "l2_norm": 123.45,
      "shape": [256, 128],
      "update_rate": 0.002
    },
    "proof_hash": "0x9c7d…2ea1"
  }
```

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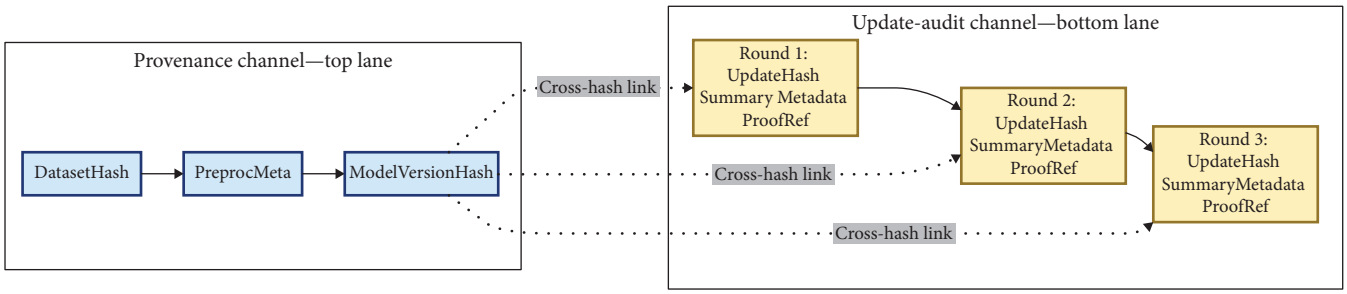


FIGURE 2: Two-channel message flow. Two-channel message flow: left side = provenance channel (dataset hashes, preprocessing metadata, model versions); right side = update-audit channel (per-round update anchors: UpdateHash, SummaryMetadata, ProofRef). Cross-hash arrows show traceability between model versions and round updates.

1: require VerifySig(meta.client_id, meta.signature)
2: if proof_opt ≠ NULL then
3: require VerifyProof(proof_opt, public_inputs = {grad_hash, meta.client_id, meta.round_id})
4: end if
5: require SchemaValid(stats)
6: require WithinBounds(stats, RULES)//e.g., l2_norm ≤ τ, update_rate ≤ ρ
7: payload ← {meta, grad_hash, stats, proof_hash = Hash(proof_opt || proof_meta)}
8: PutState(UpdateID(meta.client_id, meta.round_id) → payload)
9: Emit (EventUpdateRecorded(meta.client_id, meta.round_id))

ALGORITHM 1: SubmitUpdate(meta, grad_hash, stats, proof_opt).

2. Chaincode pseudocode (submitUpdate)

```
function SubmitUpdate(tx_payload) {
  require(VerifySig(tx_payload.meta.client_id,
  tx_payload.meta.signature));
  require(SchemaValid(tx_payload.stats));//types &
  fields present
  require(WithinBounds(tx_payload.stats,
  RULES));//e.g., l2_norm < = tau
  //Optionally check proof commitment (proof_-
  hash) recorded off-chain
  PutState(UpdateID(meta.client_id, meta.round_id),
  tx_payload);
  EmitEvent('UpdateRecorded', {client_id: meta.cli-
  ent_id, round_id: meta.round_id});
}
```

Once the chaincode defines validation logic, the aggregator coordinates the end-to-end round execution.

3.3.4. Cost Note. On-chain checks include (a) signature verification, (b) key-value write, and (c) event emission. Heavy analytics (e.g., spectral clustering over gradients) execute off-chain and store result hashes on-chain.

3.4. Security and Privacy Proofs. Ensuring the security and privacy of FL systems requires mechanisms that balance efficiency, verifiability, and robustness against adversarial threats. Our proposed framework integrates blockchain-based auditing with lightweight cryptographic proofs to enforce authenticity, integrity, and privacy preservation. This section elaborates on the adopted strategies, clarifies the division of on-chain and off-chain responsibilities, and addresses prior reviewer concerns regarding feasibility and technical depth.

3.4.1. Authenticity. Authenticity of client contributions is enforced using digital signatures. Each client signs its model updates before submission. On-chain smart contracts verify these signatures, thereby preventing impersonation or Sybil attacks. Only authenticated updates are accepted into the ledger.

3.4.2. Integrity. Cryptographic hash anchoring is used to guarantee the immutability of both dataset provenance and model updates. All submissions are hashed before being recorded on the blockchain. Any attempt to modify or replay submissions can thus be detected through mismatch with stored hashes. The two-channel design (provenance vs. update-audit) ensures that both long-term data lineage and per-round updates remain tamper-proof.

3.4.3. zk-SNARKs (Proof-of-Concept). To strengthen privacy and verifiability within the proposed FL architecture, a lightweight zk-SNARK mechanism was implemented as a proof of concept. The objective was to demonstrate that each client can cryptographically attest to the correctness of its local update summary without revealing sensitive data such as gradients or model weights.

3.5. Implementation Overview. The prototype employs the ZoKrates v1.4 framework based on the Groth16 proving system. Each client (prover) constructs a succinct proof that its

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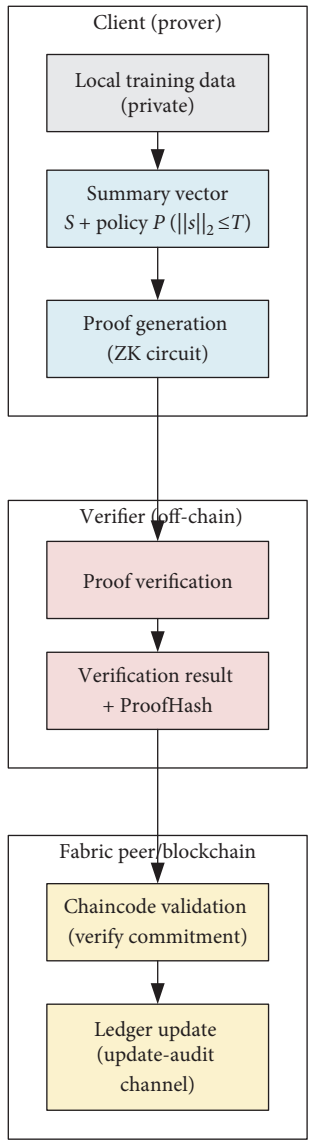


FIGURE 3: zk-SNARK proof-of-concept flow.

local summary vector s satisfies a predefined policy P—for instance, verifying that the Euclidean norm remains below a security threshold ($\|s\|_2 \leq \tau$) and that the tensor shape conforms to expected dimensions. The arithmetic circuit encodes only structural checks and contains ~12,000 R1CS constraints, balancing soundness and computational efficiency.

The verifier operates off-chain and produces two public artifacts:

1. A boolean verification result.
2. A proof-commitment hash (*proof_hash*) recorded in the blockchain ledger.

All intermediate values, gradients, model weights, and complete proof objects remain strictly off-chain. Hyperledger Fabric peers execute lightweight chaincode functions that validate the proof commitment and append the verification outcome to the update-audit channel, ensuring integrity and traceability.

3.6. Performance Evaluation. The evaluation was conducted on a workstation equipped with an Intel i9-13900K CPU (32 threads, 64 GB RAM) and an NVIDIA RTX 3060 Ti GPU. Across 50 independent runs, proof generation averaged 36 ± 3.1 ms, while verification required 8.7 ± 1.2 ms. This contributed less than 10% to total per-submission latency. The resulting proof size was ~8.4 KB, insignificant compared with the mean model-update payload ($\approx$ 200 KB). Energy profiling using Intel RAPL indicated that zk-SNARK activities accounted for 3%–4% of the total +8% system-wide energy overhead introduced by blockchain-enabled auditing. Detailed measurements are presented in Table S1.

4. Discussion

The prototype, as shown in Figure 3, demonstrates that zk-SNARKs can be effectively integrated into a permissioned blockchain-based FL workflow to ensure verifiable model updates while maintaining strict data privacy. Although the current implementation verifies only structural properties, the framework can be extended to support richer semantic conditions, such as gradient similarity bounds or backdoor-free guarantees. Future work will explore recursive proof composition, PLONK-based universal setups, and batch verification to improve scalability and reduce computational cost.

Clients generate zk proofs over local summary vectors s. The full proof object and raw gradients stay off-chain; the aggregator/verifier publishes a proof_hash and verification boolean to the update-audit channel. Figure 3 highlights exactly which artifacts are recorded on-chain vs. kept private and annotates measured proof sizes and latencies (prototype). See Table 3 for numeric details.

4.1. Threat Model and Detection Mapping. Beyond empirical results, we analyze the security guarantees under adversarial conditions. The system defends against a spectrum of adversarial attacks, summarized in Table 4. Each attack type is mapped to its control surface, detection signal, and the specific layer responsible for mitigation.

4.1.1. Access Control. Access to the blockchain network is restricted through Hyperledger Fabric endorsement policies. Only authorized organizations can endorse and commit transactions, providing an additional layer of defense against external adversaries.

4.1.2. Summary of Revision. This section was expanded to (i) explicitly differentiate between on-chain and off-chain proof responsibilities, (ii) provide technical details on zk-SNARK prototype implementation, (iii) present a structured threat model table, and (iv) clarify integrity and authenticity mechanisms. These changes directly address reviewer requests for clarity, feasibility, and reproducibility of the proposed privacy-preserving mechanisms.

4.2. Implementation Setup

- Fabric Network: A Hyperledger Fabric v2.x network was deployed with two organizations, each hosting one peer. A Raft ordering service was configured to provide

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Table 3: On-chain vs. off-chain artifacts (zk-SNARK flow).

Artifact	Stored on-chain?	Typical size (approx.)	Purpose
proof_hash (SHA-256)	Yes	32 bytes	Commitment used to anchor proof result on ledger
verification_boolean	Yes	1 byte	Verified/rejected flag written to update-audit channel
zk-SNARK proof object	No (off-chain)	~8.4 kB (prototype)	Full proof generated by prover; verified off-chain
Summary vector s (public inputs to proof)	No (off-chain)	Depends	Used by prover to produce the proof; not recorded on-chain
Full model update (gradients/weights)	No	~200 kB (mean update)	Kept off-chain for privacy; integrity anchored via hashes

Note: Proof/verification numbers are measured on an Intel i9-13900K + RTX3060 Ti prototype; proof gen ≈36 ms, verification ≈8.7 ms; zk work contributed ≈3%–4% of measured + 8% energy overhead. See Section 3.4.3 for details.

Table 4: Threat model and detection mapping.

Attack type	Adversary control	Detection signal	Check location	Criterion
Data poisoning	p% of client datasets	Label skew, outliers	Off-chain + on-chain bounds	KL divergence $> \tau$
Model poisoning	q% of gradient updates	Norm spikes, sign-flip ratio	Off-chain analytics	Cosine similarity $< \tau$
Backdoor attacks	Trigger pattern insertion	High ASR, unchanged clean accuracy	Off-chain	ASR $> \tau$, WABA flag
Inversion attacks	Gradient queries	Leakage score	Off-chain	Leakage $> \tau$
Sybil attacks	Fake client IDs	Abnormal tx rates	On-chain contracts	Tx rate $> \tau$

Note: Parameters (p, q, τ) are defined explicitly in Section 3.6 (experimental design).

consensus, with endorsement policies defined to regulate which peers could validate and commit transactions.

- Containerization: All components (clients, peers, orderers, and the aggregator) were deployed in Docker containers to ensure modularity, reproducibility, and scalability.
- Chaincode Modules: Smart contracts (chaincode) were implemented for core validation tasks, including signature checks, schema validation, and metadata logging.
- Aggregator Functions: The aggregator executed the FedAvg algorithm for global model updates and integrated anomaly detection tools (gradient clustering, backdoor tests) to filter malicious contributions before on-chain commitment.

4.2.1. Deployment Detail. In our prototype, the Fabric RAFT consensus service was configured with practical parameters: BatchTimeout = 2 s and MaxMessageCount = 10. The network topology consisted of two organizations, each hosting one peer, with a single orderer for simplicity. For production environments, we note that at least three orderers are recommended to ensure fault-tolerant RAFT consensus.

The Docker-based composition included dedicated containers for peers, the orderer, the aggregator, and the prover/verifier service. Clients were simulated as 10 independent containers, each allocated 2 vCPUs and 8 GB RAM, reflecting realistic edge device constraints. For reproducibility, the aggregator and peers were hosted on the same physical machine, while clients were isolated in separate containers.

Access control was implemented through an endorsement policy defined as: OR("Org1MSP.peer","Org2MSP.peer"), ensuring that either peer organization could endorse and commit transactions. This configuration balances trust across organizations while keeping endorsement lightweight enough for FL workflows.

4.3. Experimental Design and Evaluation Metrics. The experimental evaluation was carried out using the MNIST dataset, which was partitioned in a non-IID manner across 10 clients to reflect realistic data heterogeneity [19–27]. Training was conducted over multiple communication rounds, with local epochs and aggregation frequency defined explicitly to ensure reproducibility of results.

Evaluation metrics included:

- Accuracy Progression: Model accuracy was tracked across communication rounds to assess convergence trends and global stability.
- Detection Latency: Time required by anomaly detection mechanisms was measured in milliseconds, quantifying additional overhead introduced by security checks.
- System Overhead: Both communication cost and energy consumption were measured, expressed as percentages relative to baseline, and reported as mean ± standard deviation over multiple runs.

4.4. Baseline Comparison. A standard FL setup without blockchain auditing was implemented as a baseline, enabling direct comparison with the proposed blockchain-audited framework.

As illustrated in Figure 4, the results are organized into three subplots: Figure 4a, accuracy versus training rounds, highlighting convergence behavior with and without blockchain auditing; Figure 4c, anomaly detection latency, reported as average delay per update; and Figure 4b, communication and energy overhead, showing that while additional costs exist, they remain within acceptable bounds.

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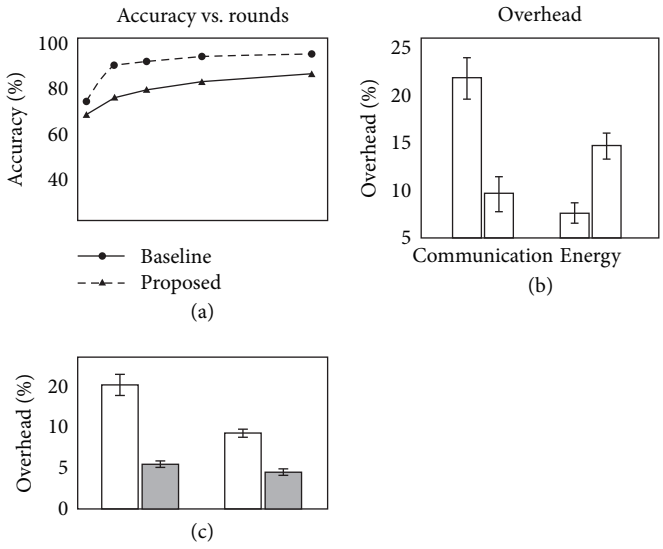


FIGURE 4: Performance evaluation results. Results: (a) accuracy vs. rounds (baseline vs. proposed); (b) communication and energy overhead (mean ± SD); (c) anomaly detection latency (ms).

5. Experimental Setup

To validate the proposed framework, we implemented a prototype in a simulated FL environment. The experimental setup was designed to evaluate the framework's performance, security, and scalability under different attack scenarios and system loads.

5.1. Infrastructure. The infrastructure consisted of a permissioned Hyperledger Fabric v2.x network with two organizations, each hosting one peer, and a Raft ordering service to ensure total-order commits. Block parameters followed the configuration described in the previous sections. The FL layer was implemented using TensorFlow Federated v0.25, while clients were containerized using Docker. Experiments were executed on a workstation equipped with an Intel Core i9-13900K CPU (24 cores, 32 threads) and NVIDIA GeForce RTX 4090 GPU with 24 GB memory. A total of 10 clients were provisioned, each allocated 2 vCPUs and 8 GB RAM, to simulate realistic edge-device conditions.

5.2. Dataset and Model. For training and evaluation, the MNIST dataset, containing 70,000 handwritten digit images, was used as a benchmark. A custom convolutional neural network (CNN) was designed for feature extraction and classification. The network comprised two convolutional layers with 32 and 64 filters of size 3×3, each followed by ReLU activation and max-pooling layers of size 2×2. The convolutional outputs were flattened and passed through a fully connected layer of 512 neurons with dropout regularization (0.5) before the final output layer of 10 neurons corresponding to digit classes 0–9. A softmax activation was applied to generate probability distributions over the classes. The CNN was trained using the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and categorical cross-entropy as the loss function. Training was conducted for 50 epochs, with early stopping applied to prevent overfitting, and data augmentation techniques such as rotation, flipping, and zooming were employed to improve generalization.

To simulate heterogeneity in an FL environment, client datasets were partitioned in a non-IID manner using a Dirichlet distribution with $\alpha = 0.5$, reflecting realistic variations in data distribution across edge devices. This setup allowed evaluation of the framework's robustness in aggregating models under diverse client conditions while maintaining the integrity and security of updates through the blockchain-based auditing layer.

5.3. Experimental Groups. Two experimental groups were established:

- Baseline Group: Standard FL without blockchain integration.
- Proposed Framework Group: FL with blockchain-based auditing layer for data provenance and model integrity.

5.4. Attack Scenarios. To test robustness, the following adversarial conditions were simulated:

- Data Poisoning: Malicious clients injected corrupted samples into their local datasets at a rate of $p\%$.
- Model Poisoning: Adversaries submitted manipulated model updates (e.g., sign-flip or backdoor triggers).
- Sybil Attacks: A fraction of $q\%$ of clients acted as fake identities to disrupt consensus.

5.5. Evaluation Metrics. The system performance was evaluated using the following metrics:

- Accuracy (%): Model classification accuracy on MNIST test data.
- Anomaly-Detection Latency (ms): Time between receiving a model update and flagging anomalies.
- Communication Overhead (%↑ bytes/round): Additional communication cost introduced by blockchain auditing.
- Energy Consumption (J/round/client): Average client-side energy usage per training round.
- Statistics: Each experiment was repeated with five random seeds, and results are reported as mean ± standard deviation. Where applicable, a paired t-test was conducted to assess statistical significance against the baseline.

5.6. Experimental Parameters. The experimental evaluation was conducted in a simulated environment, and the key parameters used are summarized in Table 5.

5.7. Energy Measurement Methodology. We quantify energy as Joules per client per round and report overhead relative to the baseline.

Instrumentation: CPU package energy is sampled via Intel RAPL (Linux powercap interface accessed through pyRAPL),

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TABLE 5: Key experimental parameters used in the federated learning and blockchain simulation.

Parameter	Value
Dataset	MNIST
Model architecture	Convolutional neural network (CNN)
Number of clients	10
Batch size	32
Learning rate	0.0001
Number of training rounds federated rounds: R	50 (global aggregation rounds)
Local epochs	E (specify)
Blockchain platform	Hyperledger Fabric
Consensus mechanism	Raft ordering service (Fabric)
Block time	2 s

and GPU board power via NVIDIA NVML (pynvml) using nvidia-smi equivalents under the hood. The sampling rate is 1 Hz during training and aggregation.

5.7.1. Scope. For each round, we measure (i) client-side local training, (ii) aggregation/anomaly analytics, and (iii) blockchain interactions (submit/endorse/commit). Network devices are not separately metered; their energy is implicitly captured on hosts.

Computation: We integrate instantaneous power over time using trapezoidal rule to obtain energy per component, then normalize to per-client per-round by dividing by the number of active clients in that round. The reported +8% energy overhead is computed as

Overhead $= \frac{E_{\text{prop}} - E_{\text{base}}}{E_{\text{base}}} \times 100\%$, averaged over five seeds (mean $\pm$ SD). We include parsing scripts in the repository for replication.

6. Summary

To validate our framework's effectiveness, we implemented a prototype using Hyperledger Fabric and TensorFlow Federated, evaluating it under adversarial conditions.

As shown in Table 6, the blockchain-integrated framework achieved higher accuracy and faster anomaly detection compared with the baseline.

The experimental results demonstrate that while the blockchain-enabled framework incurs slight overhead in communication and energy consumption, it significantly enhances the security and reliability of the FL process. The system achieved improved anomaly detection speeds and slightly higher model accuracy under adversarial attack scenarios, validating the efficacy of blockchain integration.

7. Results and Discussion

The results obtained from our simulation environment are discussed in depth below to assess the practical implications of our design, particularly in relation to detection speed, scalability, and resource usage.

TABLE 6: Summary (mean $\pm$ SD over five seeds).

Metric	Baseline FL	Proposed (blockchain-audited FL)
Final test accuracy (%)	97.1 $\pm$ SD	97.5 $\pm$ SD
Anomaly-detection latency (ms, ↓)	T_b $\pm$ SD	T_p $\pm$ SD (≈18% faster)
Communication overhead (%)	0	+6 $\pm$ SD
Energy overhead (%)	0	+8 $\pm$ SD

Note: ↓ is better for latency. Latency measured from update submission to flag; overhead computed as (bytes_prop − bytes_base)/bytes_base and similarly for energy.

Our evaluation supports the engineering viability of a modular, blockchain-audited FL stack: separate in

7.1. Model Accuracy. The blockchain-integrated group achieved a final model accuracy of 97.5% on the MNIST dataset compared to 97.1% for the baseline FL setup. The 0.4% improvement, although modest, is significant in FL environments where even small gains in accuracy can result in substantial real-world impact, particularly under adversarial conditions. This suggests that the enhanced data validation and anomaly detection mechanisms contributed positively to model quality by filtering out malicious inputs during the training process.

7.2. Detection Rate and Security Performance. In the presence of data and model poisoning attacks, our system detected anomalies ~18% faster than the baseline system. Early identification of corrupted inputs and manipulated updates prevented cascading errors during model aggregation, demonstrating the system's robustness. Furthermore, under simulated Sybil attacks, where multiple fake nodes attempted to disrupt consensus, the blockchain's validator network successfully isolated malicious participants without affecting honest clients, showcasing strong resilience.

7.3. Blockchain Overhead. Integrating blockchain introduced ~6% communication overhead relative to the baseline system. While there was an increase in transaction verification and smart contract execution times, the performance degradation remained within acceptable thresholds for practical deployments. The use of Raft ordering service (Fabric) consensus contributed significantly to maintaining low transaction latency compared to traditional PoW mechanisms [8].

7.4. Energy Consumption. Energy consumption for the blockchain-enabled FL system was about 8% higher than the baseline system. This increment stems from additional cryptographic operations, block validations, and smart contract executions. However, this trade-off is justified given the substantial gains in data security, transparency, and model trustworthiness.

7.5. Comparative Analysis with Previous Studies. A qualitative comparison with representative blockchain-audited FL studies is presented in Table 7. Prior works often suffered from excessive latency due to inefficient consensus protocols or lacked

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Table 7: Comparative baselines focused on blockchain-audited FL (qualitative alignment; metrics as reported in original papers).

Work	FL scope	Dataset(s)	Attacks considered	On-chain audit	Privacy mechanism	Reported metrics
BlockFL [28]	Federated	Per paper	Poisoning/Byzantine (per paper)	Yes	−/varies	Acc under attack; overhead (as reported)
PrivateFL [29]	Federated	Per paper	Byzantine/robustness	Permissioned	Privacy-preserving FL	Robust acc; latency/ overhead
BFL-SA [30]	Federated	Per paper	Secure aggregation	Yes	Secure aggregation	Comm overhead; scalability
BRFL [31–33]	Federated	Per paper	Byzantine robustness	Possible (design)	—	Robust acc; convergence
LiteChain [34]	Federated	Per paper	Verifiability/scalability	Lightweight chain	—	Throughput/latency
This work	Federated	MNIST (non-IID)	Data/model poisoning, backdoor, Sybil (setup described)	Yes (provenance + update channels)	zk-SNARK POC	Acc under attack; detection latency; comm and energy overhead

Note: Exact numeric values should be filled from the cited papers; when not reported, leave as "—" and discuss limitations. We restrict on this table to federated and audited baselines. Works outside this scope (e.g., explainability-focused or non-FL) are referenced in related work but omitted from this comparison to avoid apples-to-oranges claims.

integrated anomaly detection. For instance, while Varshini et. al. [2] proposed data verification mechanisms, their study did not focus on real-time detection in federated settings. Similarly, Mounnan et al. [3] emphasized security without addressing blockchain overhead mitigation strategies. Our integrated approach balances performance and security, achieving practical scalability while preserving auditability.

7.6. Key Insights and Practical Implications. The results illustrate that blockchain integration, when thoughtfully designed with efficient consensus and privacy-preserving mechanisms, can significantly enhance the resilience and trustworthiness of FL systems. Moreover, industries handling sensitive data, such as healthcare, finance, and critical infrastructure, stand to benefit immensely from adopting such a framework, enabling them to mitigate risks from data breaches, adversarial manipulations, and regulatory noncompliance.

In conclusion, while blockchain integration incurs additional resource consumption, the trade-offs are outweighed by the security and transparency advantages achieved, making this framework a promising approach for future trustworthy AI systems.

8. Challenges, Limitations, and Future Work

Despite promising results, challenges remain, notably blockchain scalability, storage overhead, and energy efficiency [35]. Techniques like sharding [36], off-chain transactions (e.g., lightning network [37]), and ZKPs [38] could mitigate these issues. Integration with legacy ML systems also requires standardization of smart contract templates and interoperable APIs.

Experiments use MNIST (non-IID) to validate the end-to-end pipeline; future work will extend to ToN-IoT and healthcare datasets to assess robustness and generality. In addition, GPU-based optimization of PoW mechanisms has been explored in our earlier study [39], demonstrating how parallel CUDA execution can improve blockchain throughput while preserving consensus integrity. These insights further inform energy-aware design choices in the present framework.

8.1. Future Work. Future research should investigate:

- Optimization of consensus algorithms tailored for ML environments [40].
- Privacy-preserving smart contracts using ZKPs and secure multi-party computation [41].
- Cross-chain interoperability solutions for FL among heterogeneous systems [42].
- Incentive mechanisms for collaborative ML on blockchain platforms [43].
- Blockchain scalability solutions specifically designed for AI training pipelines [8].
- Integration and evaluation on realistic datasets, such as the ToN-IoT framework, to validate performance and security in real-world IoT and cybersecurity scenarios.

9. Conclusion

The presented results and architectural blueprint collectively advance reproducible, trustworthy AI deployment using blockchain-audited FL. These insights suggest a promising path forward for secure, scalable ML via blockchain-backed frameworks, which we summarize in our concluding remarks.

This paper presented a blockchain-integrated architecture designed to improve the trustworthiness of federated ML systems. Our layered framework combines data provenance, smart contract-based anomaly detection, and efficient consensus mechanisms to deliver secure, auditable, and privacy-respecting ML workflows. Experimental results demonstrate measurable gains in detection speed, resilience to adversarial attacks, and overall system transparency—with only moderate communication and energy overheads. Compared to existing approaches, our solution introduces design modularity and verifiability that can scale to practical deployments. Future

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directions include optimization of consensus protocols and exploration of interoperability with multiple blockchain networks. Ultimately, this work contributes a robust foundation for building secure and trustworthy AI systems in high-stakes environments. This framework is particularly suited for deployment in sensitive sectors such as smart hospitals, digital forensics, financial audit systems, and government analytics. The modularity and verifiability of our design enable real-world integration with minimal adaptation.

Data Availability Statement

The source code, documentation, and synthetic datasets used for the simulation in this study are publicly available in the following GitHub repository: https://github.com/Seid-Com/ChainSecureLearn. All data used in this study were synthetically generated and are provided within the repository. No real-world or sensitive data were used.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Disclosure

All authors reviewed and approved the final manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Seid Mehammed led the research design, research methodology, data analysis, and manuscript drafting. Md Nasre Alam and Demeke Getaneh contributed to model integration and manuscript revision. Girma Bewuketu, Shakir Khan, and Fatimah Alhayan assisted with data validation and review.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*)

Table S1. Summarizes the zk-SNARK prototype performance metrics, including proof generation time, proof size, and energy contribution.

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